

Industrial Functional Programming ¹

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Contents

1 Supervisors

2 Applications

Supervisors

- Managing and monitoring processes
- Supervise workers and other supervisors
- Restarting processes when necessary
- Terminates processes in a predefined order

Supervisors

```
supervisor:start_link(SrvName, CModName, Args)
supervisor:start(SrvName, CModName, Args)
supervisor:start_link(CModName, Args)
supervisor:start(CModName, Args)
```

- **Calls:** `CModName:init(Args)`
- **Returns:** `{ok, {SupSpec, ChildSpecList}}`
- **SupSpec:** `{RestartStrategy, AllowedRestarts, MaxSec}`
- **RestartStrategy:** `one_for_one, one_for_all, rest_for_one`

Supervisors: ChildSpecList

- `{ok, {SupSpec, ChildSpecList}}`
- **ChildSpec:** `{Id, {Mod, Fun, Args}, Restart, Shutdown, Type, ModList}`
- `{Mod, Fun, Args} -> start_link`
- **Restart:** `transient, temporary, permanent`
- **Shutdown:** `millisec, infinity, brutal_kill`
- **Type:** `worker, supervisor`
- **Helper function:** `check_childspecs/1`

Supervisors: Dynamic children

- **Starting:** `supervisor:start_child/2`
- **Terminating:** `supervisor:terminate_child/2`
- **Restarting:** `supervisor:restart_child/2`
- **Deleting:** `supervisor:delete_child/2`

Applications

- requiring a directory structure
- handling configurations and dependencies:
`app_name.app`
- creating environment variables and configuration
- starting and stopping applications:
`application:start(AppName)`
- **Application Monitor:** `appmon:start()`

Applications Description File

```
app_name.app, app_name.app.src
```

- {application, ApplicationName, Properties}
Properties = [{Kev, Value}]
 - {description, "AppDescString"}
 - {vsn, "VsnString"}
 - {modules, ModList}
 - {registered, RegNameList}
 - {env, [{Key, Val}]}
 - {maxT, MilliSec}
 - {applications, AppNameList}
 - {mod, {Mod, Name, Args}}

Applications Behaviour

```
-module(my_app) .  
-behaviour(application) .  
-export([start/2, stop/1, ...]) .
```

```
start(normal, _Args) ->  
    ...
```

```
stop(_State) ->  
    ...
```

Starting and Stopping the Application

```
application:start(my_app) .
```

```
application:stop(my_app) .
```

On the Next Lecture ...

- NIFs
- Ports